SYMPLECTIC HYPERSURFACE EXCISION OBSTRUCTS GEOMETRIC BOUNDEDNESS

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Allowed standard prerequisites (editorial).

- Symplectic cutting, holomorphic-curve compactness and standard toric/Fukaya background as used by author. Original Sikorav, STVFluxShape and FOOO/ChoOh bibliography retained; actual source-local arguments included.

Source reference. Original reference 1.22: algebra-geometry/sources/sigma-2026-066/native/sigma26-066.tex, source lines 381–381. Later application or generalization mentioned in the author narrative; not an input to the retained target proof.

Source reference. Original reference 3.10: algebra-geometry/sources/sigma-2026-066/native/sigma26-066.tex, source lines 998–998. Later application or generalization mentioned in the author narrative; not an input to the retained target proof.

Source reference. Original reference 1.16: algebra-geometry/sources/sigma-2026-066/native/sigma26-066.tex, source lines 306–306. Later application or generalization mentioned in the author narrative; not an input to the retained target proof.

0.1. **The results.** As our first proxy for completeness, we use the well-established notion of *geometric boundedness* (see, e.g., [14, Section 4.2]): an ω-compatible almost complex structure J on M is called *geometrically bounded* if the associated Riemannian metric $g_J(\cdot,\cdot) = \omega(\cdot, J\cdot)$ is complete and has uniformly bounded sectional curvature and uniformly positive injectivity radius.[1]

We begin with an existence question.

Question 0.1. Given ω, does there exist a geometrically bounded ω-compatible almost complex structure J on M?

It is customary to call (M,ω) itself geometrically bounded if the answer to Question 0.1 is affirmative. We consider this question when we start with a geometrically bounded M and excise a subset $N \subset M$. Question 0.1 can be understood as the question: *can the distance to N of any compact subset of $M \setminus N$ be made arbitrarily large by adjusting the almost complex structure?*

Theorem 0.2. *Let M be a geometrically bounded symplectic manifold with finitely generated spherical homology, and suppose ω is rational. Let $N \subset M$ be a properly embedded symplectic hypersurface. Then $M \setminus N$ is not geometrically bounded.*

1. Proofs

1.1. **Energy confinement for J-holomorphic curves.** Throughout, if $u\colon (\Sigma, \partial\Sigma) \to (M,\omega)$ is a J-holomorphic curve with (possibly empty) boundary, its energy is $E(u) := \int_\Sigma u^*\omega \geq 0$.

The following confinement lemmas are stated for ω-compatible J. At the end of this subsection we note that both hold for ω-tame J under the uniform tameness

Date: doi:10.3842/SIGMA.2026.066; CC BY 4.0.

[1] To treat ω-tame almost complex structures, one must also impose a uniform bound on the norm of ω with respect to g_J (i.e., $|\omega|_{g_J} \leq C$).

condition from the footnote to the definition of geometric boundedness on page 1 (in Section 1.1).

Lemma 1.1 (Sikorav's confinement estimate)**.** *Let (M,ω) be a symplectic manifold and let J be an ω-compatible almost complex structure such that the associated metric $g_J(\cdot,\cdot) := \omega(\cdot, J\cdot)$ is complete and has uniformly bounded sectional curvature and positive injectivity radius (i.e., J is geometrically bounded). Then there exist a constant $r_0 > 0$ depending on the geometry and a universal constant $c > 0$ such that for every connected J-holomorphic curve u one has*

$$u(\Sigma) \subset B_{r_0 \lceil E(u)/(cr_0^2)\rceil}(u(\partial\Sigma)),$$

where $B_r(\cdot)$ denotes the closed g_J-ball of radius r about a subset and $\lceil\cdot\rceil$ is the ceiling function.

Proof. See [14, Proposition 4.4.1]. □

This confinement estimate is concerned with the large scale. Namely, it contains no information about curves with energy much smaller than cr_0^2 other than that the diameter is $\leq r_0$. In some cases, e.g., in the proof of Proposition 1.22, we need an estimate for the small scale. This is taken care of by the following lemma.

Lemma 1.2 (improved confinement)**.** *In the setting of Lemma* 1.1*, with the same r_0 and c, and letting $C := 1/c$, one has*

$$u(\Sigma) \subset B_\rho(u(\partial\Sigma)), \qquad \rho := \max\big(3CE(u)/r_0, \sqrt{CE(u)}\big).$$

Proof. The argument relies on the same monotonicity inequality used for Sikorav's lemma: if a J-holomorphic curve meets the center of a ball of radius $r \leq r_0$ and the boundary of the J-holomorphic curve does not meet the interior of the ball, then the energy in that ball is at least cr^2.

If $E(u)/\big(cr_0^2\big) \leq 1$, we apply the monotonicity inequality directly: if the image contained a point at the center of a ball of radius greater than $\sqrt{E(u)/c}$ whose interior did not meet $u(\partial\Sigma)$, then monotonicity would force $E(u) \geq cr^2 > E(u)$ (since $E(u) \geq cr^2$ implies $r \leq \sqrt{E(u)/c}$). Hence $u(\Sigma)$ is contained in a ball of radius $\sqrt{E(u)/c} = \sqrt{CE(u)}$ about $u(\partial\Sigma)$.

If $E(u)/\big(cr_0^2\big) > 1$, take the maximum number n of disjoint balls of radius r_0 that the image of u can meet. Then $n \leq E(u)/\big(cr_0^2\big)$ by monotonicity, and the image is contained in a ball of radius $(2n+1)r_0 \leq r_0\big(1 + 2E(u)/\big(cr_0^2\big)\big) < 3E(u)/(cr_0) = 3CE(u)/r_0$ around $u(\partial\Sigma)$. □

For later use, we also record a localized version, originally observed in [6].

Definition 1.3. For $a > 0$, we say that the complete Riemannian metric g_J is *a-bounded at a point $p \in M$* if $\mathrm{inj}_{g_J}(p) \geq \frac{1}{a}$ and $\sec_{g_J} \leq a^2$ on $B_{1/a}(p)$. We say that g_J is *a-bounded on a set* if it is a-bounded at every point of that set.

Remark 1.4. When g_J is a-bounded, we can take $r_0 = 1/(2a)$ [12, Theorem 4.4.1 and Remark 4.4.3].

Lemma 1.5 (local confinement)**.** *Fix $a > 0$. There exist $r_0 = r_0(a) > 0$ and a universal constant $C > 0$ with the following property. Let $K \subset M$ be compact and let $R > 0$. Assume J is ω-compatible and g_J is complete and is a-bounded on the g_J-ball $B_R(K)$. Let u be a J-holomorphic curve u with $u(\partial\Sigma) \subset K$ and write*

$$\rho := \max\big(2CE(u)/r_0, \sqrt{CE(u)}\big).$$

If $\rho \leq R$, then $u(\Sigma) \subset B_R(K)$.

Proof. The proof of Lemma 1.2 relies on the monotonicity inequality which depends only on the local geometry of a complete Riemannian metric. So taking r_0 and C as in Lemma 1.2, we obtain by the same argument that $u(\Sigma) \subset B_\rho(u(\partial\Sigma))$. □

Remark 1.6. Lemmas 1.1, 1.2, and 1.5 also hold for ω-tame almost complex structures J that are geometrically bounded (respectively a-bounded on the relevant set) and satisfy the uniform tameness condition $|\omega|_{g_J} \leq C$. The constants then depend on the tameness bound as well. See Remark 3.10.

1.2. **Proof of Theorem 0.2.** We now turn to the proof of Theorem 0.2. The idea is to pick a small Lagrangian torus L linking the hypersurface N, consider the class of the obvious "meridian" holomorphic disc intersecting N once, and show that the corresponding moduli space has nonzero evaluation degree.

Fix $p \in N$. Choose a Darboux chart $\psi\colon U \to \mathbb{C}^n$ centered at p such that $\psi(p) = 0$ and, in these coordinates, $\psi(U \cap N) = \big(\{0\} \times \mathbb{C}^{n-1}\big) \cap B_r(0)$ for some $r > 0$. For sufficiently small $t > 0$, define a Lagrangian torus $L_t \subset U$ by

$$L_t := \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid |z_1| = t, |z_i| = 10t \text{ for } i = 2, \dots, n\}.$$

Let $\alpha \in \pi_2(M, L_t)$ be the relative homotopy class represented in the chart by the disc $u_0\colon (D, \partial D) \to (\mathbb{C}^n, L_t)$, $u_0(\zeta) = (t\zeta, 0, \dots, 0)$, so $E(u_0) = \int_D u_0^* \omega_{\mathrm{std}} = \pi t^2$.

For an almost complex structure J on M, denote by $\mathcal{M}_J(L_t, \alpha)$ the moduli space of J-holomorphic discs $u\colon (D, \partial D) \to (M, L_t)$ in class α with one boundary marked point, modulo reparametrization. Let $\mathrm{ev}\colon \mathcal{M}_J(L_t, \alpha) \to L_t$ be the evaluation map at the marked boundary point.

Lemma 1.7. *There exist arbitrarily small $t > 0$ such that the following holds. For a generic geometrically bounded ω-tame almost complex structure J on M, the moduli space $\mathcal{M}_J(L_t, \alpha)$ is a smooth closed manifold of dimension n, and the evaluation map $\mathrm{ev}\colon \mathcal{M}_J(L_t, \alpha) \to L_t$ has degree 1.*

Proof. Step 1: *Index and regularity.* Trivialize TM over the chart and use the standard computation for the product torus to see that the Maslov index of α equals 2. Since the disc is simple (in particular not multiply covered), for a generic choice of ω-tame J the linearized $\bar\partial_J$-operator is surjective (see, e.g., [12]), hence the regular part $\mathcal{M}_{J,reg}(L_t, \alpha)$ consisting of maps from the disc to M is a smooth manifold of the expected dimension $n + \mu(\alpha) - 2 = n$.

Step 2: *Compactness (no bubbling).* Since ω is a rational form and $H_2(M; \mathbb{Z})$ is finitely generated, the image of ω on $\pi_2(M)$ is a discrete subgroup of $\mathbb{R}$, say $\kappa\mathbb{Z}$ for some $\kappa > 0$. Thus there exists $\hbar > 0$ such that every nonconstant J-holomorphic sphere has area at least $\hbar$. Choose a sufficiently large integer m and define $t > 0$ by the relation $\pi t^2 = \kappa/m$. By choosing m large enough, we ensure both $\pi t^2 < \hbar$ and $t < r/10$, so the construction stays well inside the Darboux chart. The image of ω on the relative homotopy group $\pi_2(M, L_t)$ is generated over $\mathbb{Z}$ by the absolute periods on $\pi_2(M)$ and the areas of the basic discs spanning the factors of the torus L_t in the chart. These generators are κ, $\pi t^2 = \kappa/m$, and $100\pi t^2 = 100\kappa/m$. Consequently, the area of any relative class in $\pi_2(M, L_t)$ is an integer multiple of κ/m. Any stable map in class α has total area $\pi t^2 = \kappa/m$. It cannot contain a nonconstant sphere component because $\kappa/m < \hbar$. Moreover, any disc bubbling would split α as a sum of two or more nontrivial relative classes, each requiring a strictly positive area. Since the minimal positive area available in $\pi_2(M, L_t)$ is exactly κ/m, and the total area of α is κ/m, such a split is impossible. Thus $\mathcal{M}_{J,\mathrm{reg}}(L_t, \alpha) = \mathcal{M}_J(L_t, \alpha)$ is a closed manifold.

Step 3: *Independence of* $\deg(\mathrm{ev})$ *and normalization.* Let J_0, J_1 be geometrically bounded and ω-tame. By [6, Theorem 4.15 (a)] (or, alternatively, Theorem 1.16 (2), (3)), one can choose a generic path $\{J_s\}_{s\in[0,1]}$ from J_0 to J_1 so the

moduli space of pairs $\{(s,u) \mid u \in \mathcal{M}_{J_s}(L_t,\alpha)\}$ is contained in a compact set. Since disc bubbling is excluded as above, this gives a compact cobordism showing that deg(ev) is independent of J.

Finally, take J which agrees with the standard complex structure on the Darboux chart and is geometrically bounded on the complement of a compact set (hence geometrically bounded globally). For t sufficiently small, Lemma 1.1 forces any J-holomorphic disc in class α with boundary on L_t to remain inside the chart, where the computation reduces to the standard toric case. In that case, regularity and deg(ev) = 1 are well known; see [2, 4]. □

We can now prove Theorem 0.2.

Proof of Theorem 0.2. Suppose by contradiction that $M \setminus N$ admits a geometrically bounded ω-tame almost complex structure J, and choose $a > 0$ such that g_J is a-bounded. Fix the small parameter t so that Lemma 1.7 applies, and note that α has algebraic intersection number 1 with N (it is the meridian disc in the local model).

Set $E := E(u_0) = \pi t^2$ and $R > \max\big(2CE/r_0(a), \sqrt{CE}\big)$, with $r_0(a)$ and C from Lemma 1.5 (so that the confinement radius $\rho = \max\big(2CE/r_0, \sqrt{CE}\big) \le R$). Let $B := B_R(L_t) \subset M \setminus N$ be the closed g_J-neighborhood of L_t of radius R. Since J is geometrically bounded on $M \setminus N$, the metric g_J is complete; hence closed bounded subsets are compact, so B is compact.

Choose an ω-tame almost complex structure J' on M which agrees with J on an open neighborhood of B and is geometrically bounded on M (e.g., interpolate to a fixed geometrically bounded structure outside a compact set). By Lemma 1.7, there exists a stable J'-holomorphic disc u in class α with boundary on L_t. Its energy equals E.

Since $J' = J$ near B and g_J is a-bounded on B, Lemma 1.5 implies $u(D) \subset B$. (We use local confinement here because J' is only guaranteed to be a-bounded on B, not globally on M.) But $\alpha \cdot N = 1$ forces any representative of α to intersect N, contradicting $B \subset M \setminus N$. □

I used AI-assisted tools for editing and improving the exposition (in particular for wording, structure, and catching typographical issues). All mathematical content, results, and any remaining errors are my own responsibility.

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